

A Korean-first tiny agentic model, for tool calling on the device.

Songgot is a from-scratch tiny language model (tens of millions of parameters) for Korean tool calling and structured extraction on phones and small devices, with a Korean-first 32k tokenizer, licence-clean data, and reproducible exact-match evaluation on Kakao FunctionChat-Bench. Weights, code and paper under Apache 2.0.

[Weights on Hugging Face](#)[Code on GitHub](#)[Live demo](#)[Pocket app \(offline, in the browser\)](#)[Paper as PDF](#)[Paper as Markdown](#)**0.90**

tokens per Hangul syllable (Songgot 32k tokenizer); Needle 2: 3.47

500

Korean items, FunctionChat-Bench SingleCall, exact-match scorer

Apache 2.0

weights, tokenizer, code, data generators, paper

0 closed-model labels

licence-clean data, provenance in the paper

Hanish Kelothe, Palette (Seoul and Bengaluru). Draft of 2026-09-10. Songgot rows in Table 2 are written from the run logs as each training run finishes; nothing on this page is estimated.

Abstract

Tiny agentic models (Needle 2, 45M parameters, 14 MB, July 2026) made on-device tool calling practical on phones and microcontrollers, but the released model is English only. Songgot is a

Korean-first model in the same class: a decoder trained from scratch with a Korean-first 32k tokenizer, bilingual Korean and English pretraining on licence-clean text, and post-training on Korean tool-calling and structured-extraction data that no closed model touched. We evaluate on Kakao's FunctionChat-Bench SingleCall (500 Korean items) with a deterministic exact-match scorer that anyone can reproduce without an API key, against Needle 2, FunctionGemma-270M, Qwen3-0.6B and Qwen3.5-0.8B under identical prompts. Results for the comparators are measured and in Table 2; Songgot rows are filled from the run logs as each training run finishes (Songgot-nano on 2026-09-10). Weights, tokenizer, data generator, scorer and this paper are released under Apache 2.0.

1. Why a Korean-first tiny model

- The class exists and is growing: Needle 2 passed 52,000 downloads within weeks of release.
- On Hugging Face as of 2026-09-09 there is no Korean tool-calling model under 1B parameters; the one Korean function-calling finetune found is a 2B Gemma.
- Korean is expensive for English-first tokenizers. On the 100 FunctionChat SingleCall queries, Qwen3's 151k vocabulary spends 1.15 tokens per Hangul syllable, Gemma's 262k vocabulary 0.98, and Songgot's 32k Korean-first vocabulary 0.90 (Table 1). At a 256 to 1024 token budget shared with tool schemas, that is context.

2. Model

Llama-style decoder, hidden 512, GQA with 8 query and 2 key-value heads, SwiGLU intermediate 1408, RoPE, tied embeddings, 32k vocabulary, trained from scratch. Two sizes: Songgot-nano, 8 layers, 39M parameters, pretrained on an Apple M5 Max with MLX (320M tokens); Songgot, 12 layers, about 50M parameters, 8xH100 run on 6B tokens queued behind GPU budget. Exports to safetensors, GGUF (f16, Q8_0, Q4_K_M) and runs under llama.cpp.

Figure 3. Songgot-nano pretraining loss, Apple M5 Max, MLX bf16, from the run log



Figure 3. Songgot-nano pretraining loss on the Mac, read from the run log

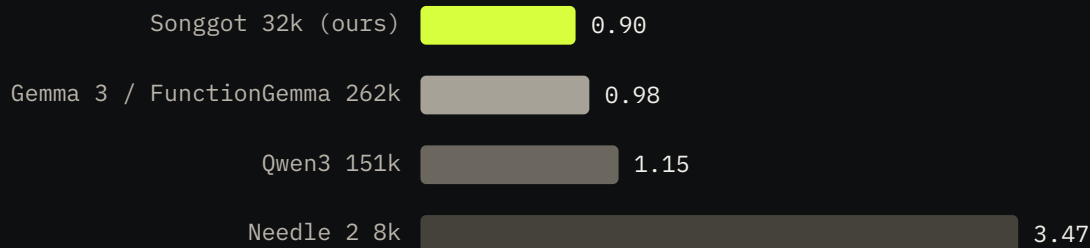
3. Tokenizer

SentencePiece BPE, 32,000 pieces, byte fallback, digits split, trained on a 300 MB sample: Korean Wikipedia 50 percent, fineweb-edu 40 percent, tool-schema JSON 10 percent. Special tokens `<|system|>` `<|user|>` `<|call|>` `<|end|>` `<|pad|>`. Post-training, evaluation and the app share one tokenizer: llama.cpp's, driven through its Python bindings with a vocabulary-only GGUF (the five specials typed as user-defined tokens). SentencePiece folds newlines into spaces and inserts a space piece before each special; llama.cpp does neither, and it is what every GGUF user runs, so the model is trained and scored on those ids.

Table 1. Tokens per Hangul syllable on the 100 FunctionChat SingleCall queries (2,071 syllables).

TOKENIZER	VOCAB	TOKENS	TOKENS PER SYLLABLE
Songgot (ours)	32k	1,873	0.90
Gemma 3 / FunctionGemma	262k	2,028	0.98
Qwen3	151k	2,392	1.15
Needle 2	8k	7,192	3.47

Figure 1. Tokens per Hangul syllable (lower is better)



100 FunctionChat SingleCall queries, 2,071 syllables; eval/tokenizer_study.py.

Figure 1. Tokens per Hangul syllable on the FunctionChat queries, lower is better

4. Data

Pretraining (all disclosed, all licence-clean):

- fineweb-edu sample-10BT (ODC-By), first 1.63B tokens under our tokenizer.
- Korean Wikipedia, dump 20231101.ko (CC BY-SA 3.0), 0.60B tokens, upsampled to roughly half of the training mix.
- No AI-Hub data (its terms forbid leaving Korea and our GPUs do not sit there), no crawled Korean web text of unclear licence.

Figure 4. Pretraining corpus under the Songgot tokenizer (billions of tokens)



Korean is upsampled to about half of each batch (p_ko 0.5); no AI-Hub data, no closed-model outputs.

Figure 4. Pretraining corpus in tokens under the Songgot tokenizer

Post-training (101,464 examples, set v3): 64,000 Korean tool calls generated from hand-written frames for 57 Korean tools (messaging, calendar, transit, home appliances, commerce, public services, health, work, information), with register variants (반말, 존댓말, 격식, 사투리, typos) and Korean date and time words resolved against a fixed calendar; 8,083 Korean calls generated by our own Palette-K-Midm (open weights) as teacher; 27,948 English calls from glaive-function-calling-v2 (Apache 2.0) over thousands of distinct tools; 6 percent "no tool applies" negatives. Each prompt carries 1, 3 to 6 or 8 tools; half of the multi-tool Korean prompts use close distractors (tools from the target's own category), mirroring the benchmark's conditions.

Tool-name style augmentation, the change that mattered most: in 52,165 rows every tool is renamed in a random style (camelCase, PascalCase, snake_case, digit or word suffixes, verb synonyms; 47,192 distinct names in the set) and the target call is renamed to match. Our first 12-layer model, trained without it, scored 0.2 percent: it copied any invented snake_case name

exactly and could not copy a single camelCase one, because every name it had seen was snake_case (the benchmark mixes both). A tiny model learns the name style it is shown; showing it every style teaches it to copy the name from the prompt. Two related fixes: glaive's arguments are single-quoted and 97 percent of its calls had failed to parse in earlier sets (the English part was 3,519 rows and mostly negatives), and "no tool" was reduced from 28 to 6 percent because the benchmark never asks for it. No closed model produced a label. A disjointness gate aborts the build if any training tool name or query appears in FunctionChat-Bench.

Two later additions (sets v4 and v5, same day): 7,745 Korean rewrites of glaive requests produced by our own Palette-K-Midm with the schemas and gold calls left untouched (Korean queries over thousands of tools instead of 57), no-parameter tools upweighted, and every no-parameter schema rendered in one of the three common JSON shapes at random. The last one came from reading failures: 45 of the benchmark's 57 no-parameter tools use `{"type":"object","properties":{},"required":[]}`, our set had almost only bare `{}`, and the model had learned the shape instead of the rule. Each set was scored before the next was built; Table 2 carries the published one and the text names the others.

v6 (2026-09-10 evening) attacks the real limit of v3-v5: those sets contain only 645 distinct real tools (the rest of their 47k names are restylings), while the benchmark's 25 tools are all new to the model. Our own teacher (Palette-K-Midm, no closed model) invents tool schemas for 60 Korean service domains in six naming styles, writes three Korean requests in different registers for each tool together with the exact call, and then, in a second pass with three distractor tools present, re-derives the call from the request alone; a pair is kept only when tool name, argument keys and values agree (99 percent agree on the name, 58 percent on everything). Three runs gave 19,293 invented tools and 36,216 verified rows; with close distractors drawn from the same domain, 30 percent of rows restyled and 5 percent turned into negatives, v6 has 152,494 rows over 113,197 distinct tool names. On the same 6B-token base, v6 lifts call accuracy from 11.4 to 23.8 percent and name accuracy from 53.6 to 63.8: the single largest step in this paper, and it came from data, not parameters or RL. A fourth synthesis run of 30 passes (v7: 100,363 verified rows over 52,483 invented tools, 219,822 rows in all, 223,319 distinct names) lifts the same base to 27.6 percent call and 69.4 name accuracy, with the close-distractor conditions moving most (4_close 13 to 22). A fifth run of 60 passes added a sibling stage: for 40 percent of invented tools the teacher writes four confusable tools in the same service (similar names, different functions), and verification shows siblings first among the distractors. v8 (336,498 synthesised rows over 183,713 tools, 467,843 rows in all, 642,439 distinct names) reaches 32.0 percent call and 71.4 name accuracy on the 6B-token base (31.0 and 72.2 on the pre3 base, a tie), with 8_close moving from 13 to 20. Three runs of the same recipe took the model from 11.4 to 32.0 percent; the curve is still rising at about four points per tripling of the synthesised set. A third epoch over v8 gives 33.0 percent (name 73.6), inside the error bar of the two-epoch run but not below it, and is the published 12-layer model. Songgot-M post-trained on the same v8 set scores 33.2 (name 73.8): with 2.5 times the parameters and the same data the two models tie, for the third time in this paper. A sixth synthesis round of 120 passes (v9: 755,011 synthesised rows over 404,424 invented tools, 907,268 rows in all) scores 33.0 percent (name 72.6): a tie with v8. The data curve, which gave about four points per tripling from v5 to v8, is flat from v8 to v9. On

this 6B-token base the synthesised data has done what it can; the remaining gap to the pretrained comparators is pretraining, and the next runs change the pretraining corpus, not the post-training set. The synthesised pool is released as [hf.co/datasets/palette-lab/songgot-tools-ko](https://huggingface.co/datasets/palette-lab/songgot-tools-ko).

Post-training recipe: full-parameter SFT, one epoch then a second at a lower rate, followed by a similarity-reward RL stage (group-relative policy optimisation against the gold call, reward = format term plus argument similarity as in STAR, Ni et al. 2026, with our own labels as the only signal). Each stage is scored on the benchmark and a stage that lowers the score is not published; Table 2 reports the published one. For the 12-layer model the RL stage tied the SFT score (11.4 percent before and after), and the published weights carry it.

What the RL stage measured. Three runs on the 12-layer model, 16 prompts x 8 samples per step, KL to the frozen SFT model. Run 1 drew prompts from the post-training set: the group mean reward was 0.97-0.99 over the first 100 steps, the eight samples of a group almost always agreed, and the benchmark did not move. Run 2 drew the same queries with every tool renamed to a name that never appears in training: reward 0.84-0.99 over 140 steps, because the model reproduces the gold call for a query it was trained on even when the tool is called something else. Run 3 used prompts the model had never seen (1,575 Korean rewrites of glaive requests held out of post-training, their renamed copies, and 3,000 held-out English rows): reward 0.70-0.85 with exact matches at 0.58-0.69, and real disagreement inside each group, the first run with a gradient worth following. After 200 of those steps the benchmark was still 11.4 percent, identical in every condition. A model this small memorises its post-training queries, so on-policy RL can only teach what SFT has not already fixed, and 200 steps of it on 3,200 novel prompts did not reach the benchmark's tools. The RL stage is kept in the recipe as a measured negative, not as a source of the published number.

5. Evaluation

Korean comparators, measured 2026-09-11 with the same prompts, scorer and each model's own chat template: Kakao's Kanana-2-1.3B-Instruct scores 70.4 percent, LG's EXAONE-4.0-1.2B 63.0. These are the Korean state of the art at on-device size and the numbers to beat; Kakao also authored FunctionChat-Bench, so Kanana's figure may reflect in-distribution training. DNA3.0-0.8B receives a prompt byte-identical to its base model Qwen3.5-0.8B (45.2 percent) and scores 4.8: it answers 421 of 500 items in prose, so its post-training removed most of the base model's tool calling. HyperCLOVA X SEED 0.5B has no tool interface in its chat template (the tools never reach the prompt), so it is listed without a score rather than with a zero.

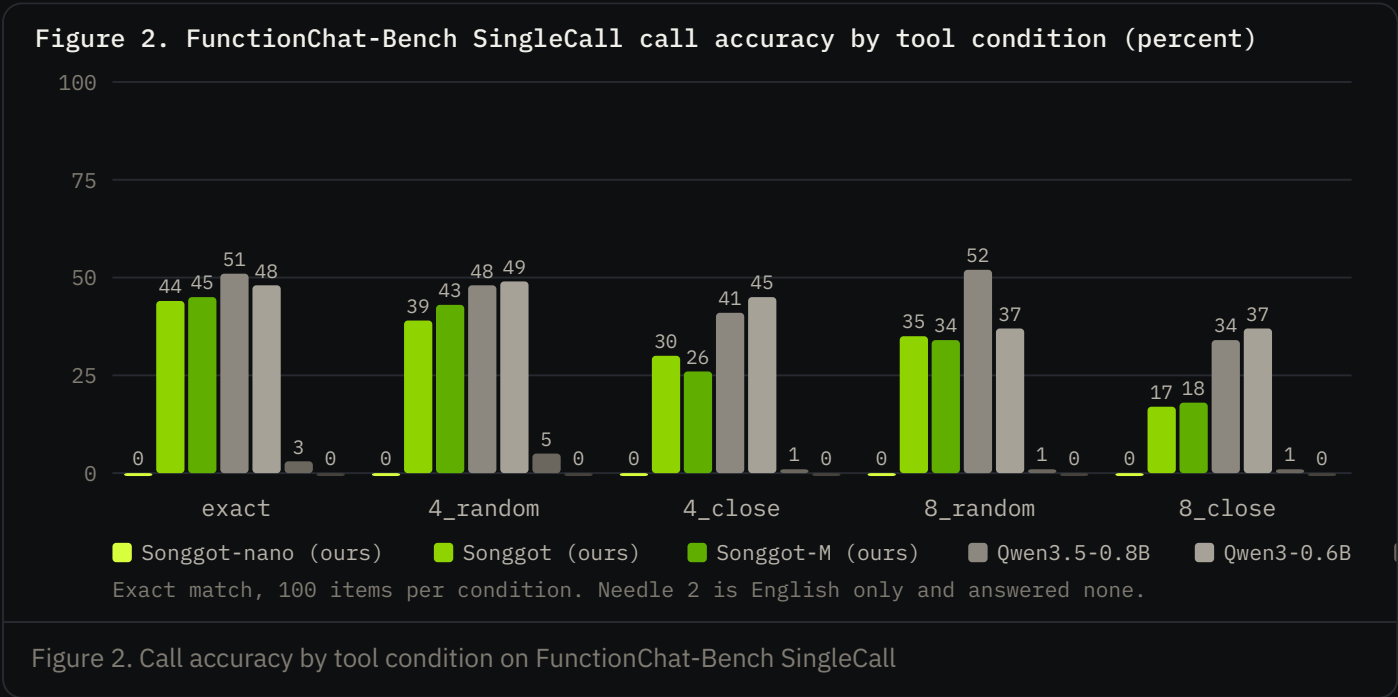
Error bars first. With 500 items the standard error of a call-accuracy figure near 25 percent is 1.9 points, so two models within about 4 points of each other are not separated by this benchmark; we treat such differences as ties and say so where they occur.

FunctionChat-Bench SingleCall (Kakao, 2024, Apache 2.0): 25 functions, 4 Korean queries each, 5 tool conditions (exact, 4 random, 4 close, 8 random, 8 close) = 500 items. The official protocol

uses GPT-4 as judge; we score with exact match on function name and arguments (numbers compared as numbers, acceptable alternatives honoured) so the number is reproducible offline. Every model receives the same tools and query, rendered in its own documented format.

Table 2. Call accuracy (exact match) on SingleCall, by tool condition, percent. Songgot rows are written by the build from the run logs; a row reads "training" until its run has finished.

MODEL	PARAMS	EXACT	4_RAN
Songgot-nano (Mac, 320M tokens)	39M	0.0	0.0
Songgot (8xH100, 6B tokens)	50M	44.0	39.0
Songgot-M (16 layers, 8xH100, 6B tokens)	126M	45.0	43.0
Needle 2	45M	0.0	0.0
FunctionGemma-270M	270M	3.0	5.0
Qwen3-0.6B	600M	48.0	49.0
Qwen3.5-0.8B	800M	51.0	48.0
Kanana-2-1.3B-Instruct (Kakao)	1.3B	76.0	72.0
EXAONE-4.0-1.2B (LG)	1.28B	73.0	65.0
DNA3.0-0.8B (Dnotitia)	0.8B	0.0	8.0
HyperCLOVA X SEED 0.5B (Naver)	0.57B	no tool-calling interface in its chat template	



6. Honest reading

- Songgot-nano, 320M pretraining tokens and one post-training epoch on the v3 set, scores 0.0 on both call and name accuracy. It answers with tool names from its own post-training catalogue (create_event, get_subway_arrival, lookup_exchange_rate) instead of the names in the prompt, and "none" for 105 of the 500 items. The 12-layer model pretrained on 6B tokens copies the right name 54 percent of the time from the same data. At 320M tokens the model has not learned to copy from context at all; the Needle 2 recipe used 200B. The nano weights stay published as the measured floor of the recipe, and the 12-layer model is the one in the app.
- Songgot-M, 16 layers and hidden 768 (126M parameters), pretrained on the same 6B tokens and post-trained on the same v5 set, scores 12.0 percent against the 12-layer model's 11.4: 32 against 26 percent with a single tool, and the same or worse with four and eight tools (13/8/7/0 against 12/10/7/2). Name accuracy is 53.0 against 53.6. Two and a half times the parameters buy three more correct calls out of 500, all of them in the easiest condition. At this token count the bottleneck is what the model has read, not how many parameters it has, and the next experiment is more pretraining tokens on the 12-layer size, not a larger model.
- More tokens of the wrong kind hurt. A second corpus (24.4B tokens: 9.0B of FineWeb-2 Korean web text, the 0.6B of Korean Wikipedia, 14.8B of fineweb-edu; same tokenizer) trained the 12-layer model for 24B tokens, four times the first run and fifteen times its Korean. Post-trained on v5 it scores 8.6 percent against 11.4 for the 6B-token model on the same data, and on v6 17.4 against 23.8. Name accuracy rose (55.8 and 66.8 against 53.6 and 63.8) while whole-call accuracy fell: the bigger corpus picks the tool a little better and fills its arguments worse, with invented arguments on tools that take none. For a tool-calling model of this size the encyclopedic and educational mix of the first corpus transfers better than general Korean web text, and the next pretraining run keeps that mix, trains longer on it, and adds a small share of tool-calling rows in the pretraining stream.
- What our data is worth on a pretrained base, as a reference we do not ship: Qwen3-0.6B (Apache 2.0) post-trained on the v5 set with its own chat template, two epochs, scores 55.4 percent against 43.2 zero-shot (name accuracy 86.6). The same v5 set gives the 50M from-scratch model 11.4, and v7 gives it 27.6. The remaining gap is what 600M parameters and trillions of pretraining tokens buy; the from-scratch line is the one we ship, and the data lever is the one that keeps moving it.
- Instruction rows in the pretraining stream help a little. A third pretraining run kept the first corpus, took three passes over it (12B tokens) and mixed in 5 percent of v6 rows rendered exactly as post-training renders them. Post-trained on v6 it scores 24.6 percent against 23.8 for the 6B-token base; post-trained on v7 it scores 25.0 against 27.6. Both differences sit inside the benchmark's error bar (section 5), so the two bases are a tie on this test, and v8 is trained on both.
- Six times the parameters on the wrong corpus lose. Songgot-L (24 layers, hidden 1024, 303M parameters) was pretrained from scratch for 24B tokens on the second corpus, the general Korean web mix that had already cost the 12-layer model six points, and post-trained on v8: 28.2 percent call, 72.4 name (published as palette-lab/songgot-l with that number). The 50M model on the first corpus scores 33.0. Size and corpus are confounded in

that comparison, so the same 303M model is being pretrained on the first corpus with the instruction bucket; until it is scored, the honest statement is that at this scale the corpus has mattered more than the parameter count every time we have measured it.

- Songgot-V after a multiple-choice stage (113,844 questions rewritten as four-way choices, 2,500 steps) scores K-DTCBench 0.258 and K-MMBench 0.294: still chance. The loss with a shuffled image equals the loss with the right one during that stage, so the model learned the answer format from the text and not from the page. Reading a page (stage 1) and reasoning over it in a benchmark format are different skills, and a 50M language model has not shown the second one. The vision line waits for a larger language model.
- The template-generated Korean data is narrow by construction; a teacher-generated set from our own Palette-K-Midm was planned and is queued behind GPU availability.
- The preview checkpoint used to open the demo on 2026-09-10 is an early one (65M tokens, 24,000 post-training examples) and is labelled as such in the table.

6b. Songgot-V, first numbers

Songgot-V pairs the from-scratch language model with an open pretrained vision tower (SigLIP2 base, 512px, Apache 2.0, trained by Google and stated as such) through a projector trained from random initialisation; the page is cut into six 512px tiles plus a thumbnail, 448 image tokens. Stage 1 aligned the projector on 245,791 Korean document pages (ko-vdr, CC BY 4.0); the loss with the right page was 3.67 against 5.54 with a shuffled page, so the model reads the image. Stage 2 trained projector and language model on 113,844 teacher-written Korean questions and answers about those pages. Scored on 2026-09-12 with next-token likelihood over the option letters, the same protocol as our baselines: K-DTCBench 0.275 (document 0.375, table 0.175, chart 0.275) and K-MMBench 0.269, against a 0.25 chance floor and 0.610 to 0.783 for the 450M to 1.7B comparators. The model had never seen a lettered multiple-choice question; a stage teaching that format is running, and these numbers stay in the paper either way.

7. Prior art and what we claim

A sweep of the Hugging Face Hub, Korean press and Korean search on 2026-09-11 (docs/PRIOR_ART_2026-09-11.md in the repository) rules out most of the sentences one would like to write, and we record them here so that nobody else has to. Korean tool calling is not new: Kakao published FunctionChat-Bench in 2024 and ships tool calling in Kanana-2-1.3B (2026-07), and LG's EXAONE-4.0-1.2B (2025-07) does agentic tool use. Tiny from-scratch Korean models are not new: KAWK-1.5-50M (51.5M, 2026-08-03), haru (6.8M, 2026-08-11) and SKT's KoGPT2 (125M, 2021) all predate this work, and KoGum-0.5B (2026-03) and KoHRM-Text-1.4B (2026-05) are from-scratch Korean models with tool-call support. Songgot is not the best Korean tool-calling model at any size: on our own scorer Qwen3.5-0.8B and Qwen3-0.6B, neither Korean-made, score 45.2 and 43.2 against our 33.0 and 33.2.

What we found nothing to contradict, and therefore claim, with its qualifiers: Songgot is the smallest openly licensed (Apache 2.0) Korean model trained from scratch for tool calling with a published FunctionChat-Bench SingleCall number, at 50M and 126M parameters, and it runs offline in a browser. Our numbers are exact-match under our own scorer, not the benchmark's GPT-4 judge, and we say so wherever they appear. Korean leaderboards (Dnotitia, Horangi, NIA K-AI) bracket models at 27B or 30B and below and have no on-device or sub-1B category, so "ranked" claims are not available to a model of this size in Korea today. For the vision model in preparation we found no Korean vision-language model under 1B parameters; we will say "we found none", not "none exists". A Naver-native sweep on 2026-09-12 (16 queries through Naver's own index, docs/NAVER_SWEEP_2026-09-12.md) found the same picture and one closed product to note: 뉴플로이 (Newploy) shipped an on-device Korean document OCR service on 2026-08-19 without weights or benchmark numbers. Kakao's Orchestration Benchmark (ICLR 2026) is a second Korean agentic benchmark on which no model under 2B parameters has been scored.

8. Release

Weights and tokenizer: hf.co/palette-lab/songgot-12l (and [songgot](https://hf.co/palette-lab/songgot-m), [songgot-m](https://hf.co/palette-lab/songgot-m)). Synthetic data: hf.co/datasets/palette-lab/songgot-tools-ko. Code, data generators, scorer, paper: github.com/hanishkeloth/songgot. Licence Apache 2.0. Korean Wikipedia attribution and CC BY-SA notice in the model card.

9. On the device

Songgot Pocket (hanishkeloth.github.io/songgot/app) is the 12-layer model running inside the browser: llama.cpp compiled to WebAssembly (wllama 3.6.1), the Q8_0 GGUF (54 MB) fetched once and kept in the browser cache, no server in the loop. It installs as a web app on iOS, Android and desktop and keeps working with the network off. The app shows the model at most six candidate tools per request (chosen by character bigram overlap with a 58-tool Korean catalogue), renders the returned call as a card, and performs the calls it can do locally (alarms, timers, notes, calendar files) while handing the rest to the right site when online. Measured on 2026-09-10 in Chrome on an Apple M5 Max, single thread: 1.5 s per request from prompt to parsed call with the 12-layer weights (755 prompt tokens, 31 generated, 21 tokens per second), 1.7 s with the 39M weights.

Questions and answers

► What is Songgot?

► **How big is it?**

► **How is it different from Needle 2?**

► **What data was used?**

► **How is it evaluated?**

► **Where are the weights and code?**

About the author

Songgot is built by **Hanish Keloth**, CTO at [Palette](#) (Seoul and Bengaluru), where he leads Palette OS, a company operating system in which governed AI agents draft the documents Korean companies run on. Palette publishes its Korean models and benchmarks in the open: Palette-K-Midm (first of seven on PALETTE-BENCH-KO v0.2, 2026-08-21), Palette-K-Doc, Palette-K-Speech and Palette Video, all at hf.co/palette-lab. Songgot continues that line at the smallest possible size.

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Songgot is released under Apache 2.0 by [Hanish Keloth](#), CTO at Palette. Korean Wikipedia text is CC BY-SA 3.0; FunctionChat-Bench is Apache 2.0 (Kakao). Numbers on this page come from the run logs in the repository; nothing is estimated. [llms.txt](#) · [paper.md](#) · [sitemap](#)